
Supplementary Materials

A Theoretical justification

Suppose we observe $\{\hat{y}_i : i = 1, \dots, n\}$ from $\hat{y}_i = y_i + \epsilon_i$, where y_i denote unobserved true values, and ϵ_i 's are i.i.d. $N(0, \sigma^2)$. Suppose $U(\hat{y}_{1:n})$ is an n -variable function of $\hat{y}_1, \dots, \hat{y}_n$. Let

$$\tilde{y}_i = \hat{y}_i - \lambda \frac{\partial U(\hat{y}_{1:n})}{\partial \hat{y}_i}.$$

Denote by $\mathbf{Y}, \hat{\mathbf{Y}}, \tilde{\mathbf{Y}} \in \mathbb{R}^n$ their vector version. Then we have the following result.

Proposition 1. *For any twice differentiable function U , the choice of $\lambda = \frac{\sigma^2 \sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial^2 U(\hat{y}_{1:n})}{\partial \hat{y}_i^2} \right\}}{\sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial U(\hat{y}_{1:n})}{\partial \hat{y}_i} \right\}^2}$ minimizes $\mathbb{E} \|\tilde{\mathbf{Y}} - \mathbf{Y}\|^2$. In particular, with this choice of λ , we have $\mathbb{E} \|\tilde{\mathbf{Y}} - \mathbf{Y}\|^2 \leq \mathbb{E} \|\hat{\mathbf{Y}} - \mathbf{Y}\|^2$. The equality holds iff $\sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial^2 U(\hat{y}_{1:n})}{\partial \hat{y}_i^2} \right\} = 0$.*

Proof. We apply the following celebrated Stein's Lemma (Csiszár and Körner, 2011):

Lemma 2 (Stein's Lemma). *Let $Z \sim N(0, 1)$. Let $h : \mathbb{R} \rightarrow \mathbb{R}$ be an absolutely continuous function (differentiable is sufficient) such that $\mathbb{E}[|h'(Z)|] < \infty$. Then $\mathbb{E}[h'(Z)] = \mathbb{E}[Zh(Z)]$.*

By Stein's Lemma, we have

$$\mathbb{E} \left[(\hat{y}_i - y_i) \left\{ \frac{\partial U(\hat{y}_{1:n})}{\partial \hat{y}_i} \right\} \right] = \sigma^2 \mathbb{E} \left\{ \frac{\partial^2 U(\hat{y}_{1:n})}{\partial \hat{y}_i^2} \right\}.$$

Hence we have

$$\begin{aligned} & \mathbb{E} \|\tilde{\mathbf{Y}} - \mathbf{Y}\|^2 - \mathbb{E} \|\hat{\mathbf{Y}} - \mathbf{Y}\|^2 \\ &= \sum_{i=1}^n \mathbb{E}(\tilde{y}_i - y_i)^2 - \sum_{i=1}^n \mathbb{E}(\hat{y}_i - y_i)^2 \\ &= \sum_{i=1}^n \mathbb{E} \left\{ \hat{y}_i - y_i - \lambda \frac{\partial U(\hat{y}_{1:n})}{\partial \hat{y}_i} \right\}^2 - \sum_{i=1}^n \mathbb{E}(\hat{y}_i - y_i)^2 \\ &= -2\lambda \sum_{i=1}^n \mathbb{E} \left[(\hat{y}_i - y_i) \left\{ \frac{\partial U(\hat{y}_{1:n})}{\partial \hat{y}_i} \right\} \right] + \lambda^2 \sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial U(\hat{y}_{1:n})}{\partial \hat{y}_i} \right\}^2 \\ &= -2\lambda \sigma^2 \sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial^2 U(\hat{y}_{1:n})}{\partial \hat{y}_i^2} \right\} + \lambda^2 \sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial U(\hat{y}_{1:n})}{\partial \hat{y}_i} \right\}^2, \end{aligned}$$

which is minimized with the choice of $\lambda = \frac{\sigma^2 \sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial^2 U(\hat{y}_{1:n})}{\partial \hat{y}_i^2} \right\}}{\sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial U(\hat{y}_{1:n})}{\partial \hat{y}_i} \right\}^2}$. Therefore, with that choice of λ , $\mathbb{E} \|\tilde{\mathbf{Y}} - \mathbf{Y}\|^2$ is also minimized, and the minimal is

$$\mathbb{E} \|\tilde{\mathbf{Y}} - \mathbf{Y}\|^2 = \mathbb{E} \|\hat{\mathbf{Y}} - \mathbf{Y}\|^2 - \frac{\sigma^4 \left[\sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial^2 U(\hat{y}_{1:n})}{\partial \hat{y}_i^2} \right\} \right]^2}{\sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial U(\hat{y}_{1:n})}{\partial \hat{y}_i} \right\}^2} = n\sigma^2 - \frac{\sigma^4 \left[\sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial^2 U(\hat{y}_{1:n})}{\partial \hat{y}_i^2} \right\} \right]^2}{\sum_{i=1}^n \mathbb{E} \left\{ \frac{\partial U(\hat{y}_{1:n})}{\partial \hat{y}_i} \right\}^2}.$$

□

B More results of toy example 1

In this supplement section, we provide additional results for toy example 1.

B.1 Results for different sample sizes

Table S1 presents experiments with a wider range of sample sizes. Specifically, we consider training set sizes (n_{train}) of 100, 300, and 1000, along with test set sizes (n_{test}) of 100, 300, and 1000, resulting in nine different combinations. Despite the increased variety of configurations, the conclusions remain consistent with those in Table 1.

Table S1: MSE and 2-WD2 for different combinations of training and evaluation dataset sizes, comparing raw and calibrated predictions. Here, ‘raw’ refers to the predictive model’s original output, ‘normalized’ refers to the predictions after normalization, and ‘cali w/ norm’ indicates calibration based on normalized predictions. The mean and standard deviation of each metric are computed from 1000 experiments.

n_{train}	n_{test}	raw		normalized		cali w/ norm	
		MSE	2-WD2	MSE	2-WD2	MSE	2-WD2
100	100	9.65 (2.43)	7.24 (2.21)	8.78 (2.26)	1.11 (0.69)	8.49 (2.19)	1.09 (0.70)
	300	9.73 (1.52)	7.22 (1.38)	8.76 (1.78)	0.84 (0.39)	8.08 (1.66)	0.76 (0.39)
	1000	9.69 (0.86)	7.14 (0.81)	8.80 (1.58)	0.74 (0.24)	8.12 (1.48)	0.44 (0.18)
300	100	8.99 (2.38)	7.19 (2.17)	6.27 (1.52)	1.04 (0.67)	6.08 (1.48)	1.01 (0.68)
	300	8.97 (1.40)	7.09 (1.25)	6.21 (1.02)	0.75 (0.34)	5.79 (0.98)	0.68 (0.34)
	1000	8.96 (0.72)	7.06 (0.66)	6.15 (0.68)	0.63 (0.22)	5.68 (0.65)	0.40 (0.17)
1000	100	8.54 (2.27)	7.04 (2.06)	5.19 (1.26)	0.88 (0.52)	5.03 (1.23)	0.86 (0.53)
	300	8.62 (1.30)	7.03 (1.18)	5.19 (0.78)	0.62 (0.29)	4.90 (0.76)	0.57 (0.30)
	1000	8.60 (0.69)	6.99 (0.63)	5.14 (0.45)	0.51 (0.17)	4.75 (0.44)	0.35 (0.14)

B.2 Normalization performance

Table S2 extends the results presented in Table 1. The comparison between ‘cali w/ norm’ and ‘cali w/o norm’ highlights that the proposed method improves predictions even without normalization, though applying normalization generally yields additional benefits.

Table S2: MSE and 2-WD2 for different combinations of training and evaluation dataset sizes, comparing raw and calibrated predictions. Here, ‘raw’ refers to the model’s original output, ‘cali w/ norm’ indicates calibration based on normalized predictions and ‘cali w/o norm’ indicates calibration based on raw predictions. The mean and standard deviation of each metric are computed from 1000 experiments.

n_{train}	n_{test}	raw		cali w/ norm		cali w/o norm	
		MSE	2-WD2	MSE	2-WD2	MSE	2-WD2
100	100	9.65 (2.43)	7.24 (2.21)	8.49 (2.19)	1.09 (0.70)	9.06 (2.47)	3.63 (1.76)
	300	9.73 (1.52)	7.22 (1.38)	8.08 (1.66)	0.76 (0.39)	8.61 (1.78)	2.73 (1.10)
	1000	9.69 (0.86)	7.14 (0.81)	8.12 (1.48)	0.44 (0.18)	8.19 (1.40)	1.90 (0.68)
300	100	8.99 (2.38)	7.19 (2.17)	6.08 (1.48)	1.01 (0.68)	7.35 (2.24)	3.78 (1.86)
	300	8.97 (1.40)	7.09 (1.25)	5.79 (0.98)	0.68 (0.34)	6.66 (1.36)	2.68 (1.13)
	1000	8.96 (0.72)	7.06 (0.66)	5.68 (0.65)	0.40 (0.17)	6.02 (0.73)	1.73 (0.55)
1000	100	8.54 (2.27)	7.04 (2.06)	5.03 (1.23)	0.86 (0.53)	6.57 (2.02)	3.64 (1.70)
	300	8.62 (1.30)	7.03 (1.18)	4.90 (0.76)	0.57 (0.30)	5.87 (1.20)	2.57 (1.07)
	1000	8.60 (0.69)	6.99 (0.63)	4.75 (0.44)	0.35 (0.14)	5.22 (0.57)	1.69 (0.51)

C More results of toy example 2

In this supplement section, we provide additional results for toy example 2.

C.1 Construction of the dynamic system

In this experiment, we constructed a nearly chaotic dynamic system to simulate complex and difficult-to-predict behaviors. The system’s state vector $\mathbf{X}_t \in \mathbb{R}^6$ evolves over time according to the following equation:

$$\mathbf{X}_t = A\mathbf{X}_{t-1} + \boldsymbol{\epsilon}_t$$

where $A \in \mathbb{R}^{6 \times 6}$ is the transition matrix that governs linear dynamics, and $\boldsymbol{\epsilon}_t \sim \mathcal{N}(0, \Sigma)$ represents Gaussian noise at each time step with zero mean and covariance matrix Σ .

To introduce near-chaotic behavior, the matrix A was sampled from a normal distribution and scaled such that the largest eigenvalue, $\lambda_{\max}(A)$, was set to 0.99. This approach introduces chaotic tendencies while maintaining stability. The covariance matrix Σ was generated independently for the training and testing datasets to enforce different steady-state distributions. Specifically, the noise covariance Σ was constructed by first generating a lower triangular matrix L with entries drawn from a normal distribution with mean 0 and standard deviation 100, then computing $\Sigma = LL^T$ to ensure positive definiteness. This procedure was carried out independently for both the training and testing datasets, resulting in distinct covariance structures for each.

C.2 Generation of datasets

The training and testing datasets were generated with sizes ranging between 100 and 1000 data points. Instead of iterating $\mathbf{X}_t$ exactly 1000 times to collect these points, we introduced two parameters: `train_interval` and `test_interval`, which control the sampling frequency (both default to 10 unless otherwise specified).

To initialize the system, the initial state $\mathbf{X}_0$ was drawn from the steady-state distribution. After simulating a long sequence of state transitions, every `train_interval` steps were selected to form the training dataset, ensuring the samples were sufficiently spaced to avoid redundancy. The test dataset was generated similarly, but using the `test_interval` to define the sampling frequency.

C.3 Knowledge marginal distribution

Our approach relies on a knowledge marginal distribution p . To estimate this distribution, we sampled 10,000 state vectors $\mathbf{X}_t$ from the steady-state distribution of the testing dataset. Each sample was then advanced by one time step to obtain $\mathbf{X}_{t+1}$, and the corresponding target values Y were computed. These 10,000 Y values were then used to fit a Gaussian Mixture Model (GMM) with 15 components, which provided an estimate of the marginal distribution of Y in the testing dataset.

C.4 Machine learning model

The goal of the experiment was to predict the norm (length) of the state vector $\mathbf{X}_{t+1}$ given $\mathbf{X}_t$. For stability and ease of data presentation, we defined the target as $Y_t = \|\mathbf{X}_{t+1}\|_2/100$. To predict Y_t based on $\mathbf{X}_t$, we employed LightGBM as the machine learning model. The model’s hyperparameters were automatically tuned using the Optuna framework. The training data was randomly split, with 80% used for training and 20% for validation, and the validation set guided the selection of the hyperparameter.

C.5 Results

The primary challenge in this experiment stemmed from the differences in steady-state distributions between the training and testing datasets, caused by the distinct covariance matrices. This mismatch led to a systematic bias in the raw predictions of the norm of $\mathbf{X}_{t+1}$, as illustrated in Figure 4. The bias manifested as a distributional shift between the predicted and true values of the testing dataset.

Similar to Table 1, the results summarized in Table S3 demonstrate a clear improvement in both prediction accuracy and distribution alignment due to the proposed calibration method. Overall, the MSE decreases by 40%-50%, indicating a significant enhancement in point-wise prediction accuracy. Furthermore, the 2-WD2, which quantifies the difference in distributions between the predicted and actual lengths of $\mathbf{X}_{t+1}$, is reduced to between one-third and one-eleventh of its original value.

Table S3: MSE and 2-WD2 for different combinations of training and evaluation dataset sizes, comparing raw and calibrated predictions. Here, ‘raw’ refers to the predictive model’s original output, and ‘cali w/ norm’ indicates calibration based on normalized predictions. The mean and standard deviation of each metric are computed from 1000 experiments.

n_{train}	n_{test}	raw		cali w/ norm	
		MSE	2-WD2	MSE	2-WD2
100	100	2191.3 (1092.5)	1832.3 (1014.3)	1207.2 (487.7)	292.8 (225.7)
	1000	2257.8 (592.8)	1916.9 (588.4)	1160.2 (308.2)	177.6 (93.4)
1000	100	1035.3 (630.9)	811.0 (558.1)	614.2 (284.7)	259.6 (193.3)
	1000	1070.8 (279.0)	861.2 (264.0)	534.3 (103.5)	158.5 (61.9)

We conducted experiments with the same settings as before when the training and testing datasets share the same Σ , i.e., the stationary distributions are identical. Comparing the results shown in Table S3 and Table S4, we can see the calibration improvement was due to its effectiveness in addressing biases due to shifts in steady state distributions. In other words, when there is no distribution shift (Table S4), the calibration does not lead to improvement.

Table S4: MSE and 2-WD2 for different combinations of training and evaluation dataset sizes, when stationary distribution remains consistent between the training and testing datasets. Here, ‘raw’ refers to the model’s original output, and ‘cali w/ norm’ indicates calibration based on normalized predictions. The mean and standard deviation of each metric are computed from 1000 experiments.

n_{train}	n_{test}	raw		cali w/ norm	
		MSE	2-WD2	MSE	2-WD2
100	100	82.11 (52.96)	30.13 (35.96)	101.54 (53.33)	33.07 (34.57)
	1000	83.83 (31.33)	26.93 (21.90)	83.59 (27.62)	9.64 (6.54)
1000	100	16.26 (7.72)	3.50 (4.20)	42.46 (35.80)	27.83 (33.55)
	1000	17.08 (4.14)	2.00 (1.95)	20.17 (5.65)	4.34 (4.36)

D Application to air-sea CO2 flux

In this appendix section, we provide more details for the application to air-sea CO2 flux example.

D.1 Dataset information

Community Earth System Model The Community Earth System Model (CESM Kay et al. (2015)) is a sophisticated climate modeling framework developed by the National Center for Atmospheric Research (NCAR). CESM is designed to simulate the interactions between the atmosphere, oceans, land surface, and sea ice, providing a comprehensive tool for studying Earth’s climate system. It integrates various components such as the atmosphere model (CAM), ocean model (POP), sea ice model (CICE), and land model (CLM), among others. CESM allows researchers to conduct experiments to understand climate variability and change, forecast future climate scenarios, and investigate the potential impacts of different factors on the global climate. Its modular structure enables flexibility and customization, making it a vital resource for climate scientists worldwide.

Surface Ocean CO2 Atlas The Surface Ocean CO2 Atlas (SOCAT Sabine et al. (2013); Bakker et al. (2016)) is an extensive, collaborative effort to compile and provide access to quality-controlled observations of carbon dioxide (CO2) concentrations in the surface ocean. SOCAT aggregates data from numerous international research programs, offering a comprehensive dataset that spans multiple decades and covers all major ocean basins. This invaluable resource supports research into the global carbon cycle, ocean acidification, and the role of oceans in mitigating climate change. By standardizing and verifying CO2 measurements, SOCAT enables accurate assessments of spatial and temporal trends in oceanic CO2 levels, facilitating better understanding and modeling of the Earth’s climate system and informing policy decisions related to carbon management and climate mitigation strategies.

D.2 Replication of SOM-FFN

For our example, we reconstructed the SOM-FFN on CESM001 by closely following the steps outlined by Gloege et al. (2021) with minor adjustments. SOM-FFN is a non-linear regression method that integrates self-organizing maps (SOM) and feed-forward neural networks (FFN) to extrapolate sparse $p\text{CO}_2$ observations onto a global $1^\circ \times 1^\circ$ grid at monthly intervals. The datasets we used include Sea Surface Temperature (SST) and Surface Chlorophyll-a (Chl-a) from satellite data, Sea Surface Salinity (SSS) from in-situ sources, Mixed Layer Depth (MLD) climatology from Argo floats, and atmospheric CO_2 mixing ratio ($x\text{CO}_2$). These variables are essential proxies for processes affecting $p\text{CO}_2$.

The initial step employed SOM to classify the global ocean into 16 biogeochemical provinces based on climatological variables, including surface ocean $p\text{CO}_2$ from Takahashi et al. (2009), SST, SSS, and MLD. This classification leverages regional consistency in the main drivers of $p\text{CO}_2$ variability. Subsequently, a non-linear regression model is developed to estimate $p\text{CO}_2$ using the environmental drivers (SST, SSS, MLD, Chl-a, and $x\text{CO}_2$). These variables vary monthly from 1982 to 2016, with any gaps filled by climatology or omitted. For each province, a separate FFN was developed based on Surface Ocean CO_2 Atlas (SOCAT) observations (Sabine et al., 2013; Bakker et al., 2016) in the first 28 years (1982-2009), avoiding mechanistic assumptions. After training, $p\text{CO}_2$ in the last 7 years (2010-2016) was reconstructed (point-wise predictions) by FFN separately in each province.

Among these 16 biogeochemical provinces, the last one primarily focuses on regions near the Arctic Ocean or Antarctica, resulting in a sampling bias due to the lack of observations in these areas. This sampling bias hampers the FFN's ability to accurately capture regional features, leading to predictions with significant errors. To address this issue, we apply a calibration technique to improve the reconstruction.

D.3 Marginal distributions of $p\text{CO}_2$ from Community Earth System Model (CESM)

In Figure S1, we present the marginal distributions of $p\text{CO}_2$ extracted from CESM 002 of the Community Earth System Model - Large Ensemble (RCP8.5) (Kay et al., 2015). These distributions are based on $p\text{CO}_2$ values from CESM002 for each month and each year from 1982 to 2016. To approximate the empirical distribution of these values, we employed a Gaussian Mixture Model with 15 components, providing a comprehensive representation of the data's variability over the specified period.

D.4 More calibration results

Figure S2 shows the distributional plot of true, raw predicted, and calibrated $p\text{CO}_2$ values in the 16th province for each month, along with the estimated density plot of $p\text{CO}_2$ values in CESM002. Although the distributions vary from month to month, our method adjusts the predicted values to make them more consistent with the true distribution.

Table S5, S6 summarize the monthly $p\text{CO}_2$ statistics for the 16th province. The "Number" column indicates the total number of data points, while the "Obv" column shows the number of distinct observations recorded each month. From the data, it is evident that the calibrated MSE values are consistently lower than the raw MSE values, indicating that the calibration process improves the accuracy of the $p\text{CO}_2$ predictions. The 2-WD2 values, which measure the distance between the predicted and true distributions, also decrease significantly after calibration.

The number of data points and distinct observations vary considerably across different months, which impacts the MSE and 2-WD2 values. For instance, months with higher numbers of observations, such as January (14146 data points, 863 observations) and June (11145 data points, 631 observations), show significant reductions in both MSE and 2-WD2 after calibration. Conversely, months with fewer observations, like April (2545 data points, 33 observations) and October (1419 data points, 8 observations), also show improvements, though the extent of change may vary due to the smaller sample sizes.

Similarly, we calculate the stability of the performance metrics for each month by LOO evaluation (see Table S7, S8). These variations highlight the robustness of the calibration method, which effectively improves prediction accuracy regardless of the number of observations or data points in different months.

Table S5: Detailed MSE for raw, normalized and calibrated pCO₂ predictions by month. The raw, normalized, and calibrated MSE values for each month in the 16th province are provided. The Number column indicates the total number of data points in the 16th province, and the Obv column shows the number of distinct observations recorded for each month.

Month	Number	Obv	Raw MSE	Normalized MSE	Calibrated MSE
Jan	14146	863	989.9	847.2	727.1
Feb	8462	408	646.3	677.3	628.2
Mar	5313	194	613.5	655.7	618.1
Apr	2545	33	911.5	895.3	835.6
May	5232	397	1185.0	426.0	405.8
Jun	11145	631	1825.0	462.9	461.9
Jul	8725	631	894.3	391.4	383.2
Aug	5095	268	924.5	274.5	247.2
Sep	3615	60	936.6	265.1	250.3
Oct	1419	8	3071.2	127.4	119.5
Nov	304	9	6402.2	251.8	232.4
Dec	9188	231	1604.5	1202.6	1095.2
Avg	75189	3733	1177.5	637.6	586.4

E Note on reproducibility

We upload a zip file (DIC.zip), which includes the code for reproducing the results in our paper, into supplementary material. Please extract the zip file and read "readme.txt" for more details. The CESM ocean pCO₂ testbed (Gloege, 2019) could be downloaded here: https://figshare.com/collections/Large_ensemble_pCO2_testbed/4568555. After downloading, please create a directory named "CESM" in the "data" folder and extract the dataset in this folder.

There are 2 python files and 4 jupyter notebooks in this folder: "model.py" is used for defining FFN and "utils.py" is used for loading CESM dataset, "toy_example_1.ipynb" is used to reproduce all the results in toy example 1, including Table 1, S1, S2 and Figure 3, "toy_example_2.ipynb" is used to reproduce all the results in toy example 2, including Table S3, S4 and Figure 4, "pco2_result.ipynb" is used to reproduce all the results in Application to air-sea CO₂ flux, including Table 2, 3, S5, S6, S7, S8 and Figure 1, 5, S1, S2. In addition, "toy_example_1.ipynb" and "toy_example_2.ipynb" could be run directly, but please run "pco2_train.ipynb" to generate necessary data before running "pco2_result.ipynb".

Table S6: Detailed 2-WD2 for raw, normalized and calibrated pCO_2 predictions by month. The raw, normalized, and calibrated 2-WD2 values for each month in the 16th province are provided. The Number column indicates the total number of data points in the 16th province, and the Obv column shows the number of distinct observations recorded for each month.

Month	Number	Obv	Raw 2-WD2	Normalized 2-WD2	Calibrated 2-WD2
Jan	14146	863	362.5	145.1	121.0
Feb	8462	408	170.7	106.2	91.8
Mar	5313	194	158.2	125.6	101.3
Apr	2545	33	389.3	91.1	66.5
May	5232	397	586.5	28.4	23.8
Jun	11145	631	1127.1	9.8	9.8
Jul	8725	631	378.5	8.7	7.2
Aug	5095	268	557.9	25.8	13.4
Sep	3615	60	542.7	46.0	43.8
Oct	1419	8	2628.1	39.9	35.2
Nov	304	9	4610.4	59.0	66.0
Dec	9188	231	644.2	172.6	136.0
Avg	75189	3733	574.4	81.8	67.0

Table S7: Mean and standard deviation of MSE from LOO evaluation for each month.

Month	Raw MSE	Normalized MSE	Calibrated MSE
Jan	990.1 (28.0)	847.4 (26.5)	727.3 (22.4)
Feb	646.7 (26.6)	677.6 (28.3)	628.5 (26.6)
Mar	613.7 (22.1)	656.0 (22.5)	618.4 (23.4)
Apr	912.3 (31.1)	895.8 (30.6)	836.3 (29.4)
May	1186.1 (49.8)	426.1 (14.1)	405.9 (13.1)
Jun	1825.4 (69.7)	463.0 (13.5)	462.0 (13.6)
Jul	895.0 (44.5)	391.7 (16.2)	383.5 (16.7)
Aug	925.3 (51.8)	274.7 (10.5)	247.4 (11.5)
Sep	937.8 (49.8)	265.4 (14.2)	250.7 (14.3)
Oct	3079.3 (151.4)	127.7 (6.7)	119.8 (6.5)
Nov	6403.9 (506.1)	253.2 (32.1)	233.8 (31.7)
Dec	1604.9 (32.0)	1202.6 (43.4)	1095.8 (37.2)
Avg	1177.9 (29.3)	637.8 (17.0)	586.7 (15.9)

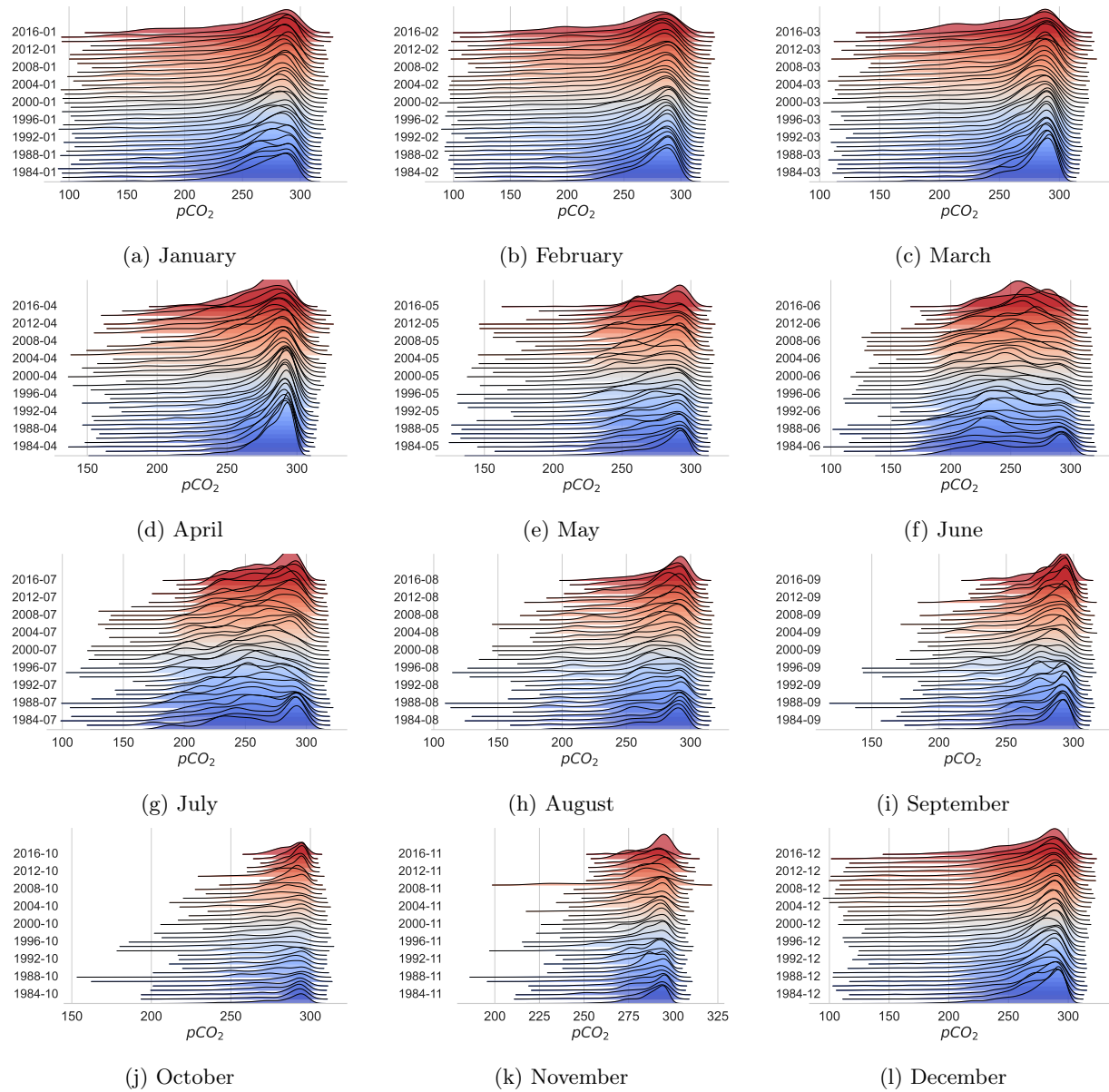


Figure S1: Marginal distributions of $p\text{CO}_2$ from CESM 002 of the Community Earth System Model – Large Ensemble (RCP8.5) (Kay et al., 2015).

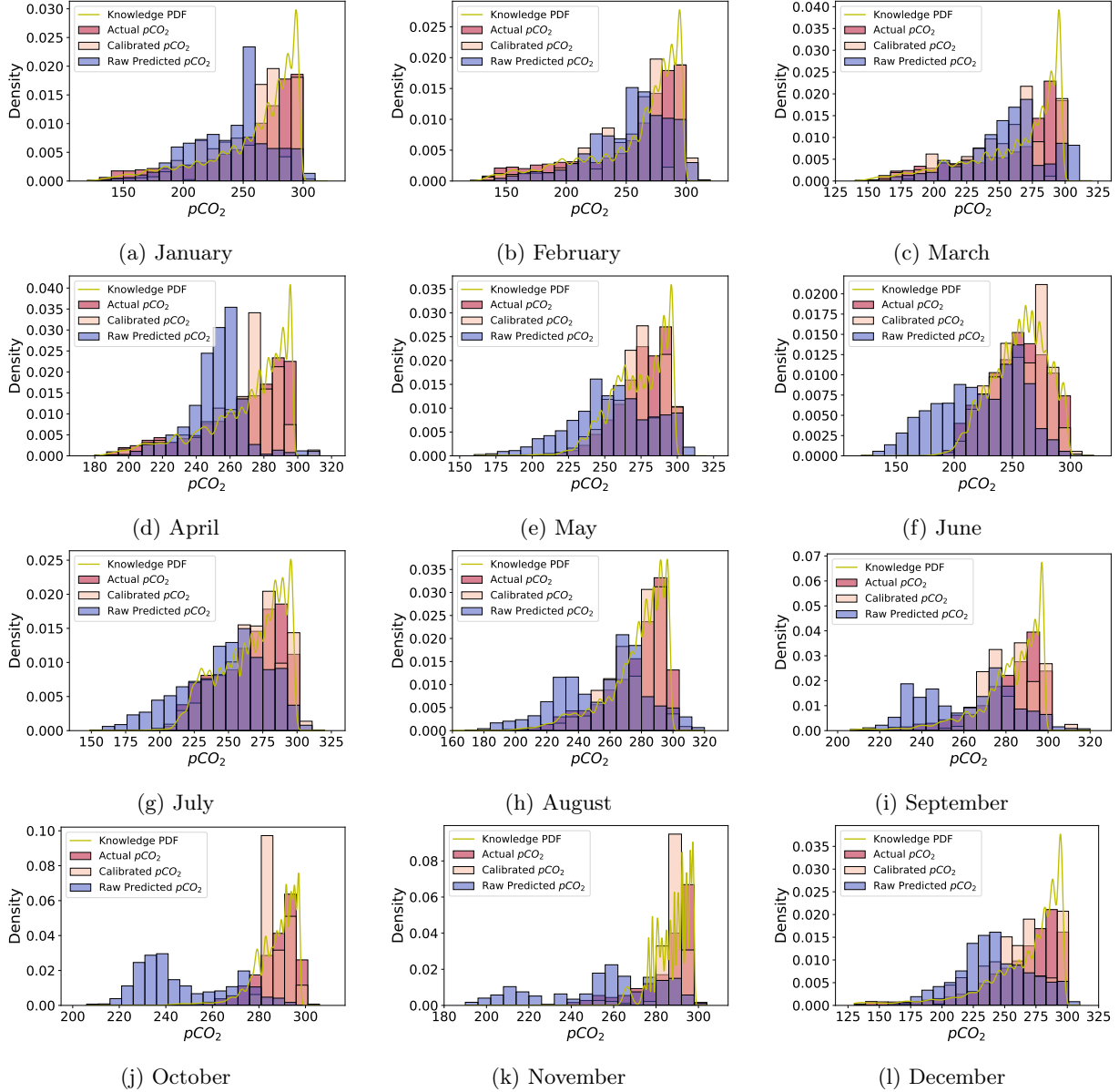


Figure S2: Calibration of $p\text{CO}_2$ predictions for the 16th province for each month: The red histogram shows the distribution of actual $p\text{CO}_2$ values. The yellow curves represent the density of a Gaussian mixture model fitted to values from CESM002. The blue histogram depicts the distribution of $p\text{CO}_2$ values predicted by the FFN. The orange histogram illustrates the distribution of $p\text{CO}_2$ values after calibration. Post-calibration, the predicted values exhibit a distribution more similar to the true values, indicating improved accuracy.

Table S8: Mean and standard deviation of 2-WD2 from LOO evaluation for each month.

Month	Raw 2-WD2	Normalized 2-WD2	Calibrated 2-WD2
Jan	363.4 (10.9)	146.1 (14.6)	121.7 (16.8)
Feb	172.0 (20.7)	107.7 (13.1)	93.2 (14.6)
Mar	159.2 (9.8)	126.7 (9.5)	102.5 (8.7)
Apr	391.3 (22.2)	93.4 (14.4)	69.0 (12.1)
May	589.3 (45.2)	29.4 (7.0)	24.7 (7.1)
Jun	1128.9 (57.7)	10.6 (3.0)	10.6 (3.0)
Jul	379.3 (31.6)	9.3 (0.9)	7.8 (0.6)
Aug	559.6 (39.2)	26.4 (1.8)	13.9 (1.7)
Sep	543.1 (27.9)	46.1 (4.0)	43.9 (4.2)
Oct	2626.1 (133.8)	40.2 (3.9)	35.4 (3.7)
Nov	4586.4 (455.2)	61.5 (20.6)	68.4 (21.9)
Dec	644.7 (29.2)	173.4 (14.4)	137.0 (11.7)
Avg	575.3 (17.7)	82.6 (4.8)	67.9 (4.8)